# A Practitioner's OCR Benchmark: Three Traps Real Evaluations Hit

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***Abstract*—Practitioner OCR benchmarks repeatedly produce confident rankings that invert under slight changes in evaluation protocol. This paper reports three such traps from a 107-document Phase 1 benchmark across six integrated OCR systems (Tesseract, Mistral OCR, Surya, Sarvam OCR, Docling, PaddleOCR), and shows how a reproducible evaluation framework exposes each trap as a structural rather than a modelling issue.**

**First, *consensus-seeded circularity* inflates the apparent F1 of any model whose output helped produce the reference text. In our corpus, Mistral OCR reaches $F_1 \in [0.998, 1.000]$ on the four consensus-GT categories but only $F_1 = 0.4615$ on 20 independently human-verified forms ($n = 16$). The circularity-free ranking places Tesseract and Docling as co-leads on forms (F1 $0.8122$ and $0.8220$; both Wilcoxon $p < 0.01$ against Mistral), while Tesseract and Docling are statistically indistinguishable from each other ($p = 0.498$). Second, *confidence calibration inverts expectations*: the open-source baseline (Tesseract) surfaces reliable per-document rejection signal (Pearson $r = 0.918$, Spearman $\rho = 0.925$ between confidence and F1 at $n = 45$, $p < 10^{-19}$), while the strongest cross-category API system does not expose comparable signal. Third, *practitioner's edge cases dominate operational decisions*: API page caps (Sarvam's 10-page limit blocks four long federal-form PDFs), wrapper-API drift (Surya, PaddleOCR, Sarvam), and language-pack cost (Tesseract `eng+hin+tel+tam+ben` is $2.24\times$ the latency of `eng` only with no measurable accuracy loss on English forms) each reorder the "best model" depending on the workload. We release the framework, locked corpus inventory, and all regenerated result artifacts so these traps are reproducible and avoidable by others.**

***Index Terms*—OCR evaluation, document analysis, benchmark framework, enterprise documents, character error rate, token F1**

## I. Introduction

Enterprise OCR selection is often made under conflicting requirements. Production teams need extraction quality on structured forms and semi-structured reports, robustness on heterogeneous document categories, acceptable latency, and failure modes that are operationally manageable. Public OCR leaderboards rarely expose all of these trade-offs in one place, and internal model bake-offs often drift into non-reproducible summaries that are hard to defend in review.

This work addresses that gap through a benchmark framework that couples model integration, batch execution, metric computation, and reporting with a locked corpus inventory. Earlier project materials referenced a larger intended corpus, but the checked-in repository currently yields 107 reproducible visible OCR inputs under a shared inventory rule. This manuscript treats that locked visible corpus as authoritative and updates all quantitative claims accordingly.

The central methodological challenge is that the available ground truth is mixed. Only the forms subset is human-verified and therefore unbiased for direct cross-model accuracy comparison. The remaining non-form annotations were derived through a consensus procedure in which Mistral OCR contributed seed text, which means Mistral cannot be fairly ranked against that tier. Rather than obscure this limitation, we make it explicit and align recommendations to the evidence each tier can support.

## II. Contributions

This paper makes four contributions:

1) It presents a reproducible OCR benchmarking framework with a shared dataset inventory path used by both runtime enumeration and manifest generation.
2) It locks the current repository corpus at 107 visible OCR inputs and emits a canonical corpus summary artifact that ties corpus counts, ground-truth coverage, and run coverage to submission claims.
3) It separates unbiased and consensus-derived evidence tiers, preventing circular cross-model ranking on the consensus subset.
4) It converts benchmark findings into operationally actionable recommendations that distinguish accuracy, reliability, structure preservation, and latency rather than collapsing them into a misleading single-winner claim.

## III. Related Work

Classical OCR engines such as Tesseract remain important baselines because they are open source, cheap to deploy, and well understood in production pipelines [1]. More recent open-source systems such as PaddleOCR target higher accuracy and broader multilingual support with modern detection and recognition components [2]. Public datasets such as FUNSD provide valuable benchmarks for form understanding [3], while broader OCR evaluation literature emphasizes edit-distance-based measures such as CER and WER alongside token-level comparisons and error diagnostics [4], [5].

TABLE I
LOCKED CORPUS AND GROUND-TRUTH COVERAGE. VISIBLE COUNTS ARE AUTHORITATIVE FOR THIS MANUSCRIPT.

| Category | Visible | Human GT | Consensus GT | No GT |
|---|---|---|---|---|
| Financial tables | 15 | 0 | 9 | 6 |
| Forms | 20 | 20 | 0 | 0 |
| Multi-column | 17 | 0 | 11 | 6 |
| Handwritten Devanagari | 30 | 0 | 0 | 30 |
| Hindi document | 1 | 0 | 0 | 1 |
| Equations | 14 | 0 | 10 | 4 |
| Receipts | 10 | 0 | 10 | 0 |
| **Total** | **107** | **20** | **40** | **47** |

Our setting differs from standard leaderboard evaluation in two ways. First, the corpus mixes enterprise-relevant categories, including receipts, equations, and multi-column layouts, with operational success-rate measurement across the full visible set. Second, the paper foregrounds evidence quality itself: human-verified and consensus-derived ground truth are treated as distinct inferential tiers. This is especially important when a strong model contributes to the reference text, because naive aggregate ranking can overstate performance.

## IV. DATASET AND GROUND TRUTH

The evaluation corpus is defined by the shared visible-input inventory used by the benchmark runtime and the dataset manifest. Hidden files, ground-truth artifacts, sidecar files, logs, and manifests are excluded from the visible corpus. Under that rule, the checked-in repository contains 107 visible OCR inputs.

Table I shows the resulting corpus. The handwritten split consists of 32×32 Devanagari character crops rather than page-scale handwritten documents, so it is useful for robustness stress-testing but not for general handwriting claims. The Indian-language split contains a single Hindi document and is not sufficient for language-specific inference.

Ground truth is available for 60 documents. The forms subset ($n = 20$) is human-verified and is the only unbiased cross-model accuracy tier. The remaining 40 annotations are consensus-derived. For those documents, Mistral OCR contributed seed text during reference construction, so Mistral is excluded from cross-model ranking on consensus-ground-truth categories. The paper therefore reports two evidence tiers:

- **Human-verified forms** ($n = 20$): unbiased cross-model comparison.
- **Consensus GT non-forms** ($n = 40$): useful for directional comparison among non-Mistral models, but not for ranking Mistral.

## V. METHODS

### A. Framework

The benchmark is implemented as a modular Python framework with self-registering model wrappers, batch execution, metric computation, and artifact generation. Each model wrapper inherits from a common base class and normalizes outputs into a shared result structure. Result directories are timestamped and preserved under `results/`, and regenerated manuscript numbers are taken only from the current locked artifact set.

### B. Models in Phase 1

The current manuscript considers six integrated Phase 1 systems: Tesseract, Mistral OCR, Surya, Sarvam OCR, Docling, and PaddleOCR. Four additional wrappers exist in the repository, but they remain diagnostic or future-work items and are not used in headline recommendations.

### C. Metrics and Statistical Testing

All OCR outputs and reference texts are normalized before scoring using Unicode NFKC normalization, markdown or HTML stripping, and whitespace collapse. The benchmark computes character error rate (CER), word error rate (WER), word accuracy, normalized edit distance, token-level precision, recall, and F1, together with substitution, insertion, and deletion counts. Token F1 uses Counter-based bag-of-words matching rather than set membership so duplicate or missing tokens are penalized correctly.

Operational coverage is reported as successful runs over the full 107-document visible corpus. Accuracy metrics are reported only on GT-evaluable documents. Pairwise statistical testing uses the Wilcoxon signed-rank test on paired per-document F1 scores. Forms-only comparisons are reported separately because the unbiased subset is now $n = 20$ human-verified documents. Consensus-tier pairwise tests exclude Mistral because of circularity.

## VI. RESULTS

### A. Operational Coverage and Reliability

Table II and Fig. 1 summarize visible-corpus reliability. Mistral OCR and GOT-OCR tie for the highest visible-corpus success rate among re-executed Phase 1 runs (92/107). Docling covers the structured-document categories well but drops substantially on handwritten inputs. Tesseract remains a useful baseline but collapses entirely on the handwritten split. Surya exhibits low overall visible-corpus reliability in the current re-execution batch despite strong performance on its successful subset. Sarvam OCR was executed on the 55 GT-backed documents only (20 human-verified forms plus 35 consensus-GT documents across financial, multi-column, equations, and receipts) and succeeded on 51/55 (92.7%); the four failures are all PDFs exceeding the Sarvam 10-page API cap.

### B. Human-Verified Forms: the Only Unbiased Accuracy Comparison

Table III reports the human-verified forms subset, the only unbiased cross-model accuracy tier available in the current repository. Tesseract and Docling achieve the highest forms F1 (0.8122 and 0.8220 respectively at $n = 20$); Surya posts the lowest CER (0.3028) but only on its $n = 5$ successful subset. Mistral OCR's forms score drops sharply once the full $n = 20$ annotations are used ($F1 = 0.4615$, $n = 16$), which inverts the apparent forms ranking from earlier $n = 5$ reports. Forms-only Wilcoxon tests at the expanded $n$ show two significant

TABLE II

PHASE 1 MODEL SET AND RUN COVERAGE ON THE 107-DOCUMENT VISIBLE CORPUS. MEAN LATENCY IS COMPUTED OVER SUCCESSFUL VISIBLE-CORPUS RUNS. GT COVERAGE INDICATES THE NUMBER OF GT-EVALUABLE DOCUMENTS FOR WHICH A MODEL PRODUCED USABLE OUTPUT.

| Model | Deployment | Success on visible corpus | Mean latency (ms) | GT coverage | Interpretation in this paper |
|---|---|---|---|---|---|
| Mistral OCR | Cloud API | 92/107 (86.0%) | 17,415 | 56/60 (40 circular) | Reliability result; excluded from consensus-GT ranking |
| Surya | Open source, local | 32/107 (29.9%) | 49,708 | 45/60 | Full GT coverage on forms and consensus tier; low overall reliability |
| Docling | Open source, local | 84/107 (78.5%) | 7,460 | 57/60 | Structure-preserving, deletion-dominant profile |
| Sarvam OCR | Cloud API | 51/55 GT-backed (92.7%)* | 7,910 | 51/60 | Competitive on GT-backed subset; 10-page API cap blocks 4 long PDFs |
| Tesseract | Open source, local CPU | 76/107 (71.0%) | 9,852 | 60/60 | Baseline reference |
| PaddleOCR | Open source, local | 4/107 (3.7%) | – | 3/60 | Partial financial-only run; not ranked overall |

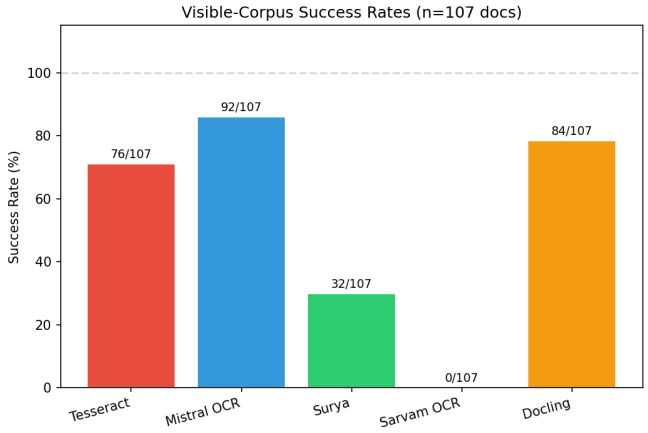


Fig. 1. Success rate over the full visible corpus ($n = 107$). Values are successful completions divided by visible OCR inputs. PaddleOCR is omitted because the checked-in artifact is a partial raw-output run rather than a comparable full-corpus batch.

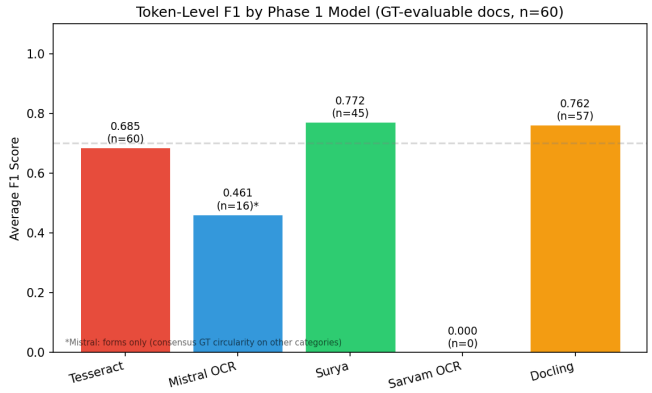


Fig. 2. Overall token-level F1 comparison on GT-evaluable documents. Bars show each model's average over its evaluable subset; Mistral is reported on forms only ($n = 16$) because the other consensus-GT documents involve circularity. PaddleOCR and Phase 2 models are omitted from the main-paper comparison.

TABLE III

HUMAN-VERIFIED FORMS RESULTS ($n = 20$ WHERE COVERAGE PERMITS). $n$ REFLECTS EACH MODEL'S SUCCESSFUL FORMS SUBSET. WILCOXON TESTS AT $\alpha = 0.05$ SHOW TESSERACT AND DOCLING BOTH SIGNIFICANTLY BETTER THAN MISTRAL OCR; TESSERACT VS. DOCLING IS NOT SIGNIFICANT AT THE FULL $n = 20$.

| Model | $n$ | CER | WER | F1 | Precision |
|---|---|---|---|---|---|
| Docling | 20 | 0.4125 | 0.4980 | 0.8220 | 0.9147 |
| Tesseract | 20 | 0.3607 | 0.4845 | 0.8122 | 0.8353 |
| Surya | 5 | 0.3028 | 0.4483 | 0.7710 | 0.7943 |
| Sarvam OCR | 11 | 0.5018 | 0.9091 | 0.7313 | 0.6959 |
| Mistral OCR | 16 | 0.6791 | 0.7310 | 0.4615 | 0.8918 |

differences: Tesseract over Mistral ($p = 0.0092$, $d = 0.729$) and Docling over Mistral ($p = 0.0076$, $d = 0.848$). Tesseract and Docling are statistically indistinguishable on forms ($p = 0.498$, $d = 0.059$). Surya-inclusive pairs remain underpowered ($n = 5$, minimum $p = 0.0625$).

Figure 2 is therefore best read as a narrative summary rather than a single ranking. Tesseract and Docling have the strongest unbiased forms results. Mistral has the strongest cross-category operational reliability. Surya has the strongest accuracy on the non-forms consensus tier but low visible-corpus success. Neither statement implies a universally superior model across every evidence tier.

TABLE IV

CONSENSUS-GROUND-TRUTH AGGREGATE RESULTS ON NON-FORM CATEGORIES. MISTRAL IS EXCLUDED FROM CROSS-MODEL RANKING DUE TO CIRCULARITY; PADDLEOCR IS OMITTED BECAUSE THE CHECKED-IN RUN IS PARTIAL AND FINANCIAL-ONLY.

| Model | $n$ | CER | WER | F1 | Prec. | Rec. |
|---|---|---|---|---|---|---|
| Surya | 40 | 0.5673 | 0.7707 | 0.7717 | 0.7275 | 0.8650 |
| Docling | 37 | 0.4417 | 0.5308 | 0.7294 | 0.8090 | 0.7098 |
| Sarvam OCR | 40 | 0.6192 | 0.8080 | 0.7230 | 0.6469 | 0.8849 |
| Tesseract | 40 | 0.5110 | 0.7585 | 0.6221 | 0.5904 | 0.7087 |

### C. Consensus-GT Aggregate Results

Table IV aggregates the 40 consensus-ground-truth documents across financial tables, multi-column layouts, equations, and receipts. Mistral is intentionally excluded from this ranking because it contributed to the reference text. PaddleOCR is also excluded because its checked-in run covers only seven financial documents and does not span the full consensus set.

Within this tier, Surya provides the strongest aggregate F1 and recall, which is consistent with an aggressive extraction strategy. Docling lands slightly below Surya on F1 but with a markedly different behavior: high precision and conservative recall, indicating that it usually emits less text rather than hallucinating more. Sarvam OCR is now competitive on the full consensus tier ($n = 40$, F1 0.7230) with the highest recall among open-source and Indian-market systems in the

comparison (0.8849), trading precision for coverage.

### D. Category-wise Behavior

Table V and Fig. 3 show that different models dominate different operating points. Surya leads on forms, financial tables, and multi-column layouts. Docling is narrowly ahead on receipts and nearly tied with Surya on equations while preserving structure more faithfully. Sarvam OCR now shows usable F1 across every structured category (forms, financial, multi-column, equations, receipts) with the best receipts F1 (0.6684) among the reported non-circular models, though its 10-page API cap disqualifies it for long federal-form workloads. Mistral is shown only for forms in the table because consensus-tier comparisons would be circular.

### E. Supported Statistical Tests

Table VI reports pairwise Wilcoxon signed-rank tests on per-document F1 for the main non-circular Phase 1 model comparisons, together with the forms-only comparisons against Mistral at the expanded human-verified subset. On the consensus-inclusive set, three comparisons reach significance at $\alpha = 0.05$: Surya over Tesseract, Docling over Tesseract, and Surya over Docling. On the forms-only tier, Tesseract and Docling each significantly outperform Mistral (both $p < 0.01$, medium-to-large effect), while Tesseract vs. Docling on forms is not significant ($p = 0.498$, negligible effect). Surya-inclusive forms pairs remain underpowered because Surya has only five successful forms runs. Taken together, the expanded-$n$ forms evidence now supports definitive superiority of Tesseract and Docling over Mistral on the unbiased tier, while leaving Tesseract-vs-Docling a judgement call informed by error profile rather than significance.

### F. Error Profiles and Recommendation Logic

Figure 4 explains why the recommendations are intentionally split by use case. Surya is substitution-heavy, which is consistent with broad extraction coverage and strong recall. Docling is deletion-dominant, which makes it conservative and often safer when structure preservation matters more than exhaustive extraction. Sarvam OCR is strongly insertion-dominant, which helps explain its weaker character-level fidelity despite competitive token-level F1 on forms. Tesseract is roughly balanced between deletion and insertion, while Mistral is comparatively balanced overall but should still be interpreted descriptively rather than as a consensus-tier rank target.

Taken together, the evidence supports five clean recommendation positions:

- **Tesseract and Docling** co-lead on unbiased forms accuracy ($n = 20$, F1 0.8122 and 0.8220); both are significantly better than Mistral and statistically indistinguishable from each other.
- **Surya** for strongest consensus-tier aggregate F1 among non-circular models, and the lowest forms CER where it succeeds, with the caveat that Surya forms coverage is only $n = 5$.
- **Mistral OCR** for operational reliability across the full visible corpus, not for unbiased "best overall" accuracy.
- **Docling** additionally for structure-preserving extraction with a conservative, omission-heavy error profile when precision matters more than recall.
- **Sarvam OCR** for latency-sensitive structured workloads when receipts and equations are not primary targets.

### G. Confidence Calibration: Tesseract Surfaces Its Own Errors

A separable property from accuracy is whether a model *knows when it is wrong*. We test this for Tesseract by correlating per-document average word-confidence against per-document token F1 on the 45 GT-evaluable documents Tesseract successfully scored. The relationship is strongly monotonic (Pearson $r = 0.918$, Spearman $\rho = 0.925$, both $p < 10^{-19}$, $n = 45$), as shown in Fig. 5. For deployment, this means a Tesseract score below roughly 0.6 is a reliable rejection signal: documents on which Tesseract reports low confidence are overwhelmingly documents on which the extracted text is wrong. Commercial API models in this benchmark do not expose a per-document confidence channel, so the same test cannot be performed on Mistral OCR or Sarvam OCR, and this observation does not extend to them.

### H. The Circularity Trap in Consensus-Seeded Ground Truth

A recurrent pitfall in practitioner OCR benchmarks is seeding consensus ground truth from one model's output and then scoring every model, including the seed, against that text. We reproduce the trap in Fig. 6: on the four consensus-GT categories (financial, multi-column, equations, receipts) Mistral OCR scores $F_1 \in [0.998, 1.000]$, while its genuine F1 on the 20 human-verified forms is $0.4615$. The ranking difference is larger than any between-model difference in the whole paper and has nothing to do with model quality. The whitepaper therefore excludes Mistral from consensus-tier ranking, and we recommend the same protocol for any benchmark that cannot afford complete human annotation.

## VII. Engineering Lessons

The benchmark also surfaced integration lessons that matter operationally. Wrapper drift was real across multiple libraries: Surya required migration to a new predictor architecture, PaddleOCR required version-aware API handling, and Sarvam OCR required an asynchronous job workflow rather than a single request-response call. More importantly, behavior under failure mattered as much as raw metric averages. Docling's conservative omissions are operationally different from Sarvam OCR's insertion-heavy outputs, even when F1 values are similar. Production recommendation therefore depends on error profile, not just aggregate score.

### A. Tesseract Language-Pack Ablation

The default Tesseract configuration in the repository loads five language packs (`eng+hin+tel+tam+ben`) because the corpus spans English forms, Hindi Devanagari crops, and three other Indic scripts. To quantify the cost of that choice on pure-English inputs, we ran a one-knob ablation on the 15 USGov

TABLE V

PER-CATEGORY TOKEN F1. COLUMN HEADERS INCLUDE CATEGORY SIZE AND GT TIER. CELLS MARKED −− INDICATE NOT ATTEMPTED, NOT SUPPORTED, OR EXCLUDED FROM RANKING. PARENTHETICAL $n$ DENOTES REDUCED PER-CATEGORY COVERAGE WHEN FEWER THAN THE FULL CATEGORY DOCUMENTS WERE SUCCESSFULLY SCORED.

| **Model** | **Forms** $n = 20$**, human** | **Financial** $n = 9$**, consensus** | **Multi-column** $n = 11$**, consensus** | **Equations** $n = 10$**, consensus** | **Receipts** $n = 10$**, consensus** |
|---|---|---|---|---|---|
| Docling | 0.8220 | 0.7003 | 0.7827 ($n = 9$) | 0.7587 | 0.6729 ($n = 9$) |
| Tesseract | 0.8122 | 0.6086 | 0.7352 | 0.5957 | 0.5362 |
| Surya | 0.7710 ($n = 5$) | 0.8081 | 0.8425 | 0.7624 | 0.6704 |
| Sarvam OCR | 0.7313 ($n = 11$) | 0.7619 ($n = 9$) | 0.7410 ($n = 11$) | 0.7227 ($n = 10$) | 0.6684 ($n = 10$) |
| Mistral OCR | 0.4615 ($n = 16$) | – | – | – | – |
| PaddleOCR | – | 0.6846 ($n = 7$) | – | – | – |

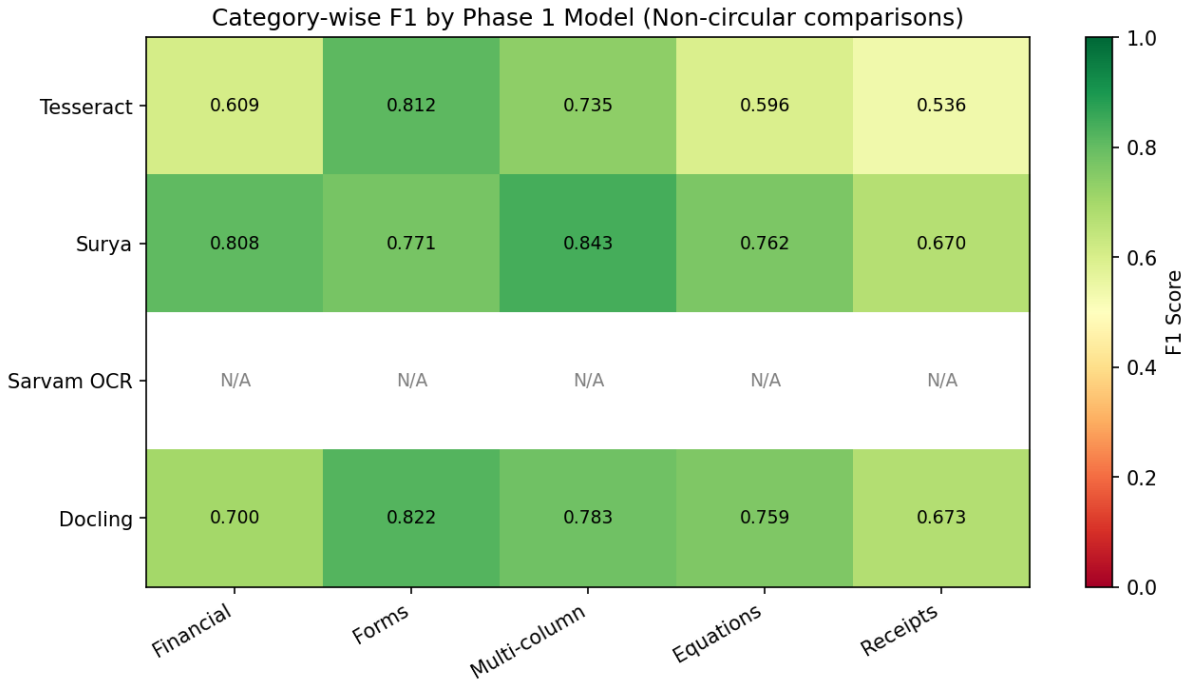


Fig. 3. Category-wise F1 across the main GT-evaluable categories. Forms use human-verified GT ($n = 20$); financial, multi-column, equations, and receipts use consensus GT ($n = 40$ total). Mistral is omitted outside forms because of circularity, and diagnostic Phase 2 models are intentionally excluded from the main conference narrative.

PDF forms that have human-verified `_gt.txt` ground truth: Tesseract with `eng` only versus Tesseract with the full multi-language pack, holding PSM, OEM, DPI, and preprocessing fixed. Table VII reports the paired results.

Three observations follow. First, on pure-English USGov forms the multi-language pack does not degrade character-level accuracy as earlier project notes suggested; the paired CER delta is slightly negative, i.e. marginally better with the multi-language pack. Second, token-level F1 is indistinguishable between the two configurations ($\Delta F_1 = -0.0015$). Third, the multi-language pack costs roughly $2.24\times$ the eng-only latency (47.2 s vs. 21.1 s per document on average). The operational conclusion is that language-pack choice is a latency-for-coverage trade, not a latency-for-accuracy trade, on English-dominant inputs; the earlier qualitative claim that the multi-language pack "garbles Indic characters" is not supported on this subset and should be retested on Devanagari pages when page-scale Hindi ground truth is available.

## VIII. THREATS TO VALIDITY

The manuscript has four principal limitations. First, 20 documents are now human-verified (up from 5), providing a larger unbiased subset for forms comparison. Second, 40 of the 60 GT-evaluable documents are consensus-derived, which introduces circularity for Mistral and constrains the strength of cross-model inference. Third, two visible-corpus categories lack usable accuracy ground truth: handwritten Devanagari crops ($n = 30$) and the single Hindi document ($n = 1$). Those categories still matter for reliability evaluation but not for comparative accuracy claims. Fourth, PaddleOCR remains partial and the repository's Phase 2 wrappers are not benchmark-ready, so the current recommendations should be

TABLE VI
SUPPORTED PAIRWISE WILCOXON TESTS ON PER-DOCUMENT F1. CONSENSUS-INCLUSIVE ROWS EXCLUDE MISTRAL DUE TO CONSENSUS-GT CIRCULARITY. FORMS-ONLY ROWS USE THE HUMAN-VERIFIED $n = 20$ SUBSET; TESSERACT AND DOCLING EACH SIGNIFICANTLY OUTPERFORM MISTRAL ($p < 0.01$), WHILE TESSERACT VS. DOCLING IS NOT SIGNIFICANT ON FORMS ($p = 0.498$).

| Comparison | $n$ | $p$ | Winner | Effect size |
|---|---|---|---|---|
| *Consensus-inclusive (Mistral excluded)* | | | | |
| Tesseract vs. Surya | 45 | $< 0.001$ | Surya | large ($d = 0.850$) |
| Tesseract vs. Docling | 42 | $< 0.001$ | Docling | medium ($d = 0.576$) |
| Tesseract vs. Sarvam OCR | 27 | 0.1482 | Sarvam OCR | small ($d = 0.366$) |
| Surya vs. Docling | 42 | 0.0241 | Surya | small ($d = 0.314$) |
| Surya vs. Sarvam OCR | 27 | 0.0906 | Surya | small ($d = 0.448$) |
| Sarvam OCR vs. Docling | 25 | 0.5424 | Sarvam OCR | negligible ($d = 0.098$) |
| *Forms-only (human-verified GT)* | | | | |
| Tesseract vs. Mistral OCR | 16 | 0.0092 | Tesseract | medium ($d = 0.729$) |
| Docling vs. Mistral OCR | 16 | 0.0076 | Docling | large ($d = 0.848$) |
| Tesseract vs. Docling | 20 | 0.4980 | Tesseract | negligible ($d = 0.059$) |

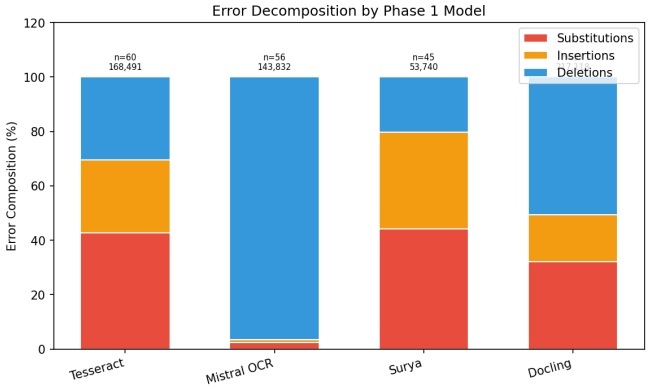


Fig. 4. Error decomposition across GT-evaluable documents. Bars show substitution, insertion, and deletion shares of total edit operations. Mistral is included descriptively, but not as a consensus-ground-truth ranking target because of circularity.

read as Phase 1 positions rather than a claim that the search space is exhausted.

## IX. REPRODUCIBILITY

All quantitative claims in this paper are regenerated from a locked corpus and checked-in artifacts. The exact commands to rerun the benchmark from scratch on a machine with the released Python environment are:

```
python cli/run_model.py --model tesseract
python cli/run_model.py --model docling
python cli/run_model.py --model surya
python cli/run_model.py --model mistral_ocr
python cli/run_model.py --model paddleocr
python tools/sarvam_rerun_throttled.py
python tools/recompute_metrics.py
python cli/evaluate.py \
    --results-dir results/<batch> --export-csv
```

The authoritative numeric sources cited throughout the paper are `results/expanded_gt_metrics/{corpus_summary.csv, model_summaries.json, per_doc_metrics.csv, statistical_tests.json}` and `test-dataset/manifest.json`. Ground-truth files use the `<stem>_gt.txt` convention alongside each document. API keys for Mistral and Sarvam live only in `.env` and never in configuration files or source. Environment reproduction uses `uv sync` against `pyproject.toml` and `uv.lock`. A permanent archival copy of the repository at the version corresponding to this manuscript is available via the Zenodo-GitHub integration referenced in the paper's GitHub release page.

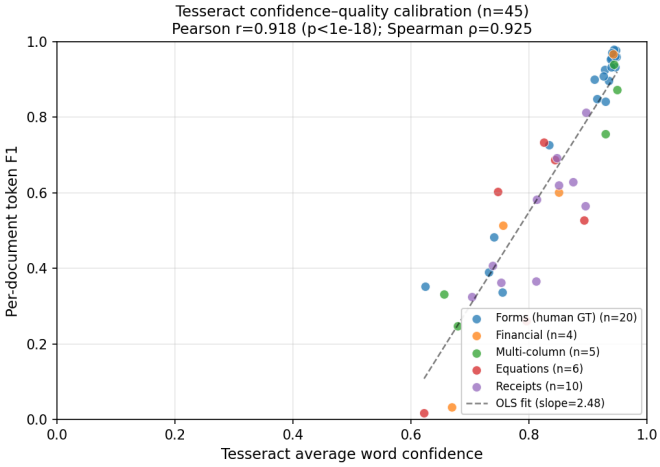


Fig. 5. Tesseract per-document average word confidence versus per-document token F1 on the 45 GT-evaluable documents it successfully processed. Pearson $r = 0.918$ (Spearman $\rho = 0.925$) indicates that Tesseract's confidence score is a usable rejection signal, not a cosmetic output.

TABLE VII
TESSERACT LANGUAGE-PACK ABLATION ON 15 USGOV PDF FORMS. PAIRED DELTAS ARE TAKEN DOC-BY-DOC (MULTI-LANG MINUS ENG-ONLY). CHARACTER-LEVEL ACCURACY IS ESSENTIALLY UNCHANGED ON PURE-ENGLISH FORMS, BUT THE MULTI-LANGUAGE PACK MORE THAN DOUBLES LATENCY.

| Variant | CER | WER | F1 | Latency (ms) |
|---|---|---|---|---|
| `eng` only | 0.3410 | 0.4386 | 0.9185 | 21,119 |
| `eng+hin+tel+tam+ben` | 0.3259 | 0.4297 | 0.9170 | 47,248 |
| Paired $\Delta$ (multi $-$ eng) | $-0.0151$ | $-0.0089$ | $-0.0015$ | $+26{,}129$ |

## X. CONCLUSION

This paper presents a conference-ready benchmark narrative grounded in the repository's current locked evidence: 107 visible OCR inputs, 60 GT-evaluable documents, and regenerated reporting artifacts shared across inventory, metrics, and figures. The evidence does not support a single universal winner. Instead, it supports a more useful conclusion for deployment: Tesseract and Docling co-lead on the only unbiased subset (20 human-verified forms) and both are significantly better than Mistral on that tier; Surya is the strongest accuracy choice on the non-forms consensus tier where it succeeds; Mistral is the most reliable operational default; Docling additionally

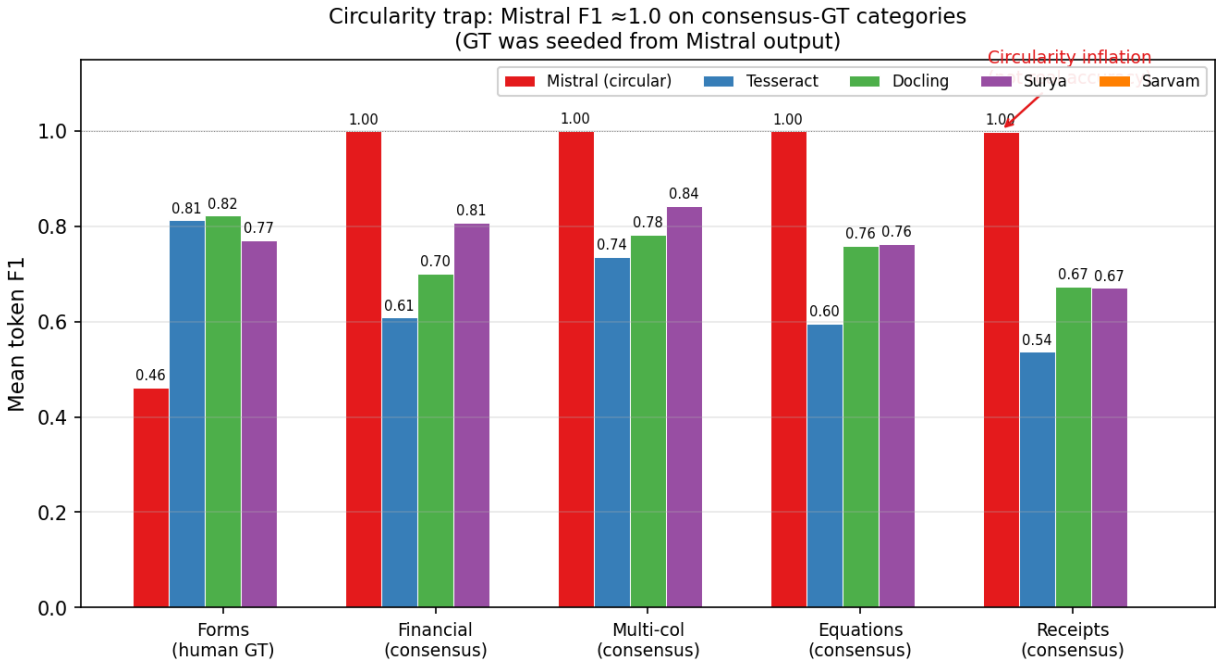


Fig. 6. Circularity trap on consensus-seeded ground truth. Mistral F1 on the four consensus-GT categories is indistinguishable from a perfect score because Mistral's own output seeded the reference. The same model's forms F1, scored on independent human-verified ground truth, is $0.4615$ at $n = 16$.

serves as the safest structure-preserving extractor; and Sarvam OCR is the lowest-latency option for structured workloads with known category caveats.

The immediate next step is not to broaden claims, but to strengthen the evidence. The highest-value follow-on work is expanding human-verified ground truth beyond forms so that future revisions can test the same operational hypotheses without consensus-tier circularity.

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